Real Banking System using Python

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INTRODUCTION

Banking systems are essential for managing financial transactions in modern society. Traditional banking involves manual processes which are time-consuming and prone to errors. With the advancement of technology, banking operations have become digital and automated.

This project presents a **Real Banking System** implemented using Python. It simulates core banking functionalities such as account creation, login authentication, deposit, withdrawal, balance inquiry, and fund transfer using a Command Line Interface (CLI).

MOTIVATION

The motivation behind this project is to understand how real banking systems work and to implement a simplified version using basic programming concepts. It helps in learning file handling, logic building, and system design.

PROBLEM STATEMENT

Manual banking systems are inefficient, lack automation, and are prone to human errors. There is a need for a system that can:

- Automate banking operations
- Store data efficiently
- Provide secure access to users

OBJECTIVES

- To develop a banking system using Python
- To implement secure login using PIN
- To perform deposit and withdrawal operations
- To enable fund transfer between accounts
- To store data using file handling

EXISTING SYSTEM

Existing systems include:

- Manual banking systems
- Digital banking apps using databases

Limitations:

- Manual systems are slow
- Complex systems require advanced tools

PROPOSED SYSTEM

The proposed system is a CLI-based banking system that:

- **Uses file handling instead of databases**
- **Provides basic banking operations**
- **Ensures data persistence**

ADVANTAGES

- **Simple and easy to use**
- **No external libraries required**
- **Efficient for small-scale systems**
- **Helps understand real-world applications**

DISADVANTAGES

- **No encryption**
- **Limited scalability**
- **No graphical interface**

SYSTEM REQUIREMENTS

Hardware:

- Computer/Laptop
- Minimum 4GB RAM

Software:

- Python 3.14
- Code Editor (VS Code / PyCharm)

SYSTEM ARCHITECTURE

User → CLI Interface → Python Logic → Text File Storage

MODULES

1. Account Creation Module
    - Creates new user account
2. Login Module
    - Authenticates user using PIN
3. Deposit Module
    - Adds money to account
4. Withdrawal Module
    - Deducts money from account
5. Balance Module
    - Displays account balance
6. Transfer Module
    - Transfers money between accounts

ALGORITHM

1. Start program
2. Display main menu
3. If user selects create account → store data
4. If login → verify credentials
5. Show banking menu
6. Perform selected operation
7. Update file
8. Repeat until exit

IMPLEMENTATION DETAILS

- File handling used for data storage
- Data stored in .txt file
- Each account stored as a record

OUTPUT

- Account creation confirmation
- Login success/failure
- Balance display
- Transaction confirmation

LIMITATIONS

- No encryption
- No database
- Limited features

CONCLUSION

The project successfully demonstrates how a banking system can be implemented using Python. It provides a clear understanding of real-world banking operations and highlights the importance of automation